Desmos Graphing Tactics, Regression & Systems Strategy

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the community theater are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?

Answer: 7. Reading the two labeled intersection points from the graph, $(x_1, y_1) = (-3, 3)$ and $(x_2, y_2) = (4, 3)$. Therefore, $x_2 - x_1 = 4 - (-3) = 7$.

2 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the mountain observatory are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?

Answer: 1. Reading the two labeled intersection points from the graph, $(-3, 1)$ and $(3, 1)$, we have $x_1 = -3$, $x_2 = 3$, and $c = 1$. Thus $x_1 + x_2 + c = -3 + 3 + 1 = 1$.

3 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the solar microgrid facility are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?

Answer: 5. Reading the two labeled intersection points from the graph, $(x_1, y_1) = (-3, 4)$ and $(x_2, y_2) = (2, 4)$. Therefore, $x_2 - x_1 = 2 - (-3) = 5$.

4 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the publishing house are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?

Answer: 4. Reading the two labeled intersection points from the graph, $(-3, 2)$ and $(5, 2)$, we have $x_1 = -3$, $x_2 = 5$, and $c = 2$. Thus $x_1 + x_2 + c = -3 + 5 + 2 = 4$.

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5 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the hydroponic greenhouse are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?

Answer: 7. Reading the two labeled intersection points from the graph, $(x_1, y_1) = (-3, 5)$ and $(x_2, y_2) = (4, 5)$. Therefore, $x_2 - x_1 = 4 - (-3) = 7$.

6 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the robotics club are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?

Answer: 3. Reading the two labeled intersection points from the graph, $(-3, 3)$ and $(3, 3)$, we have $x_1 = -3$, $x_2 = 3$, and $c = 3$. Thus $x_1 + x_2 + c = -3 + 3 + 3 = 3$.

7 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the wildlife sanctuary are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?

Answer: 5. Reading the two labeled intersection points from the graph, $(x_1, y_1) = (-3, 1)$ and $(x_2, y_2) = (2, 1)$. Therefore, $x_2 - x_1 = 2 - (-3) = 5$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the robotics club, consider the quadratic function $p(t) = 12t^2 - 13$ over the closed interval $[13, 28]$. What is the average rate of change $\Delta p/\Delta t$ on $[13, 28]$?

Answer: 492. The average rate of change is $\frac{p(28) - p(13)}{15} = 12((28) + 13) = 492$.

9 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the meteorological station, let the functions h and k be defined by $h(u) = 22u - 7$ and $k(u) = u^2 + 3$. What is the composite value $h(k(10))$?

Answer: 2259. First compute $k(10) = 10^2 + 3 = 103$. Then $h(103) = 22(103) - 7 = 2259$.

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10 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the urban bike-share station, suppose in a hypothetical 50-question section, a student answers 90% of the first 30 questions correctly, then 60% of the remaining 20 correctly. How many total questions did the student answer correctly?

Answer: 39. 0.90(30)=27 correct, and 0.60(20)=12 correct, for a total of 27+12=39.

11 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the sports complex, consider the quadratic function $h(u) = 8u^2 - 15$ over the closed interval $[3, 22]$. What is the average rate of change $\Delta h/\Delta u$ on $[3, 22]$?

Answer: 200. The average rate of change is $\frac{h(22) - h(3)}{19} = 8((22) + 3) = 200$.

12 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the wind energy cooperative, since $4^3 = 64$ and $5^3 = 125$, a problem asks which integer choice is closest to $\sqrt[3]{100}$. Choices: A) 5, B) 4, C) 10, D) 20.

Answer: 5. $\sqrt[3]{100} \approx 4.64$, which is closer to 5 (a difference of 0.36) than to 4 (a difference of 0.64), so 5 is the best integer choice.

13 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the astronomical society, in the xy-plane, the graph of $y = (x - 2)^2 - 16$ at the municipal library intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

Answer: 8. Setting $y = 0$ gives $(x - 2)^2 = 16 \Rightarrow x - 2 = \pm 4$, so the x-intercepts are -2 and 6. The distance between them is $(6) - (-2) = 8$.

14 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the film archive, in the xy-plane, a circle with equation $(x - 7)^2 + (y - 9)^2 = 64$ represents a boundary at the mountain observatory. What is the greatest value of b for which the horizontal line $y = b$ is tangent to the circle?

Answer: 17. The circle has center $(7, 9)$ and radius $r = 8$. The highest point on the circle is $(7, 9 + 8) = (7, 17)$, so the greatest value of b for which $y = b$ is tangent to the circle is 17.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

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15 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the regional airport shuttle, in the xy-plane for the water treatment plant model, the parabola $y = x^2 + 6x + k$ and the horizontal line $y = 6$ intersect at exactly one point. What is the value of the constant k?

Answer: 15. Setting $x^2 + 6x + k = 6$ gives $x^2 + 6x + (k - 6) = 0$. For the graphs to intersect at exactly one point, the discriminant must equal zero: $(6)^2 - 4(1)(k - 6) = 0 \Rightarrow k - 6 = 9 \Rightarrow k = 15$.

16 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane for the botanical conservatory model, the parabola $y = x^2 + 4x + k$ and the horizontal line $y = 3$ intersect at exactly one point. What is the value of the constant k?

Answer: 7. Setting $x^2 + 4x + k = 3$ gives $x^2 + 4x + (k - 3) = 0$. For the graphs to intersect at exactly one point, the discriminant must equal zero: $(4)^2 - 4(1)(k - 3) = 0 \Rightarrow k - 3 = 4 \Rightarrow k = 7$.

17 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the graph of $y = (x - 2)^2 - 16$ at the municipal library intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

Answer: 8. Setting $y = 0$ gives $(x - 2)^2 = 16 \Rightarrow x - 2 = \pm 4$, so the x-intercepts are -2 and 6. The distance between them is $(6) - (-2) = 8$.

18 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, a circle with equation $(x - 4)^2 + (y - 6)^2 = 25$ represents a boundary at the community theater. What is the greatest value of b for which the horizontal line $y = b$ is tangent to the circle?

Answer: 11. The circle has center $(4, 6)$ and radius $r = 5$. The highest point on the circle is $(4, 6 + 5) = (4, 11)$, so the greatest value of b for which $y = b$ is tangent to the circle is 11.

19 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane for the regional bakery model, the parabola $y = x^2 + 10x + k$ and the horizontal line $y = 6$ intersect at exactly one point. What is the value of the constant k?

Answer: 31. Setting $x^2 + 10x + k = 6$ gives $x^2 + 10x + (k - 6) = 0$. For the graphs to intersect at exactly one point, the discriminant must equal zero: $(10)^2 - 4(1)(k - 6) = 0 \Rightarrow k - 6 = 25 \Rightarrow k = 31$.

20 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the graph of $y = (x - 5)^2 - 49$ at the coastal aquarium intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

Answer: 14. Setting $y = 0$ gives $(x - 5)^2 = 49 \Rightarrow x - 5 = \pm 7$, so the x-intercepts are -2 and 12. The distance between them is $(12) - (-2) = 14$.

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Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the botanical conservatory are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?

a. 3 ✓

b. 5

c. 2

d. 6

Reading the two labeled intersection points from the graph, $(x_1, y_1) = (-1, 1)$ and $(x_2, y_2) = (2, 1)$. Therefore, $x_2 - x_1 = 2 - (-1) = 3$.

22 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the municipal library are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?

a. 5

b. 6

c. 2

d. 3 ✓

Reading the two labeled intersection points from the graph, $(-2, 2)$ and $(3, 2)$, we have $x_1 = -2$, $x_2 = 3$, and $c = 2$. Thus $x_1 + x_2 + c = -2 + 3 + 2 = 3$.

23 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the regional bakery are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?

a. 7

b. 8 ✓

c. 10

d. 5

Reading the two labeled intersection points from the graph, $(-1, 4)$ and $(5, 4)$, we have $x_1 = -1$, $x_2 = 5$, and $c = 4$. Thus $x_1 + x_2 + c = -1 + 5 + 4 = 8$.

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24 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the coastal aquarium are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?

a. 4 ✓

b. 6

c. 3

d. 7

Reading the two labeled intersection points from the graph, $(x_1, y_1) = (-2, 5)$ and $(x_2, y_2) = (2, 5)$. Therefore, $x_2 - x_1 = 2 - (-2) = 4$.

25 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the science museum are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?

a. 4

b. 8

c. 5 ✓

d. 7

Reading the two labeled intersection points from the graph, $(x_1, y_1) = (-1, 2)$ and $(x_2, y_2) = (4, 2)$. Therefore, $x_2 - x_1 = 4 - (-1) = 5$.

26 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the urban bike-share station are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?

a. 5

b. 6 ✓

c. 8

d. 10

Reading the two labeled intersection points from the graph, $(-2, 3)$ and $(5, 3)$, we have $x_1 = -2$, $x_2 = 5$, and $c = 3$. Thus $x_1 + x_2 + c = -2 + 5 + 3 = 6$.

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27 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the ceramic studio are shown in the xy-plane with their two points of intersection (x_1, c) and (x_2, c) labeled. What is the value of $x_1 + x_2 + c$?

a. 9

b. 3

c. 6

d. 7 ✓

Reading the two labeled intersection points from the graph, $(-1, 5)$ and $(3, 5)$, we have $x_1 = -1$, $x_2 = 3$, and $c = 5$. Thus $x_1 + x_2 + c = -1 + 3 + 5 = 7$.

28 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of a quadratic function $y = f(x)$ and a horizontal line $y = g(x)$ modeling two trajectories at the agricultural cooperative are shown in the xy-plane with their two points of intersection (x_1, y_1) and (x_2, y_2) labeled, where $x_1 < x_2$. What is the distance $x_2 - x_1$ between the x-coordinates of the intersection points?

a. 5

b. 9

c. 6 ✓

d. 8

Reading the two labeled intersection points from the graph, $(x_1, y_1) = (-2, 1)$ and $(x_2, y_2) = (4, 1)$. Therefore, $x_2 - x_1 = 4 - (-2) = 6$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the art gallery, let the functions p and q be defined by $p(t) = 22t - 9$ and $q(t) = t^2 + 11$. What is the composite value $p(q(10))$?

a. 2444

b. 2424

c. 2455

d. 2433 ✓

First compute $q(10) = 10^2 + 11 = 111$. Then $p(111) = 22(111) - 9 = 2433$.

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30 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the culinary institute, in the coordinate plane, the graph of $y = h(u)$ is translated 5 units right and 6 units up to form $y = k(u)$, mapping $X(2, 50)$ to Y. What is the sum of coordinates of Y on $y = k(u)$?

a. 63 ✓

b. 74

c. 58

d. 69

The image point is $Y(7, 56)$, whose coordinates sum to 63.

31 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the water treatment plant, let the functions u and v be defined by $u(s) = 16s - 12$ and $v(s) = s^2 + 6$. What is the composite value $u(v(14))$?

a. 3236

b. 3220 ✓

c. 3226

d. 3208

First compute $v(14) = 14^2 + 6 = 202$. Then $u(202) = 16(202) - 12 = 3220$.

32 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the sports complex, in the coordinate plane, the graph of $y = p(t)$ is translated 2 units right and 15 units up to form $y = q(t)$, mapping $P(20, 67)$ to Q. What is the sum of coordinates of Q on $y = q(t)$?

a. 104 ✓

b. 119

c. 117

d. 102

The image point is $Q(22, 82)$, whose coordinates sum to 104.

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33 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the maple syrup farm, suppose a student plans to spend exactly 1.5 minutes on each of 30 questions in a 50-minute section. How many total minutes would that use?

a. 50 minutes exactly

b. **45 minutes, leaving a 5-minute buffer ✓**

c. 40 minutes

d. 42 minutes

1.5 × 30 = 45 minutes, leaving 50-45=5 minutes as a buffer.

34 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the planetarium, suppose a student answers the first 25 of 30 questions at about 1.5 minutes each, in a 50-minute section. How many minutes are left for the remaining 5 questions?

a. 15 minutes

b. 10 minutes

c. **12.5 minutes ✓**

d. 7.5 minutes

25 × 1.5 = 37.5 minutes used; 50 - 37.5 = 12.5 minutes remain.

35 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the regional airport shuttle, a function f satisfies the identity $f(x-19)=22x+7$ for all real x. What is the value $f(16)$?

a. 799

b. 758

c. 784

d. 777 ✓

Setting $x-19=16 \Rightarrow x=35$ gives $f(16)=22(35)+7=777$.

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36 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the wind energy cooperative, suppose a student is 5 questions into a 55-minute, 27-question section after using 12 minutes, with a target pace of about 2 minutes per question. Are they on pace, ahead, or behind?

a. Behind pace (5 questions should take about 10 minutes, but 12 were used) ✓

b. Ahead of pace

c. Exactly on pace

d. Cannot be determined

At 2 minutes/question, 5 questions should take 10 minutes; using 12 minutes means the student is running 2 minutes behind.

37 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the marine research vessel, suppose a student has 20 minutes left and 8 questions remaining. What's the average time available per remaining question?

a. 2 minutes

b. 2.5 minutes ✓

c. 3 minutes

d. 1.5 minutes

20 minutes / 8 questions = 2.5 minutes per question.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the maple syrup farm, in the xy-plane, the graph of $y = (x - 5)^2 - 16$ at the culinary institute intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

a. 13

b. 8 ✓

c. 4

d. 16

Setting $y = 0$ gives $(x - 5)^2 = 16 \Rightarrow x - 5 = \pm 4$, so the x-intercepts are 1 and 9. The distance between them is $(9) - (1) = 8$.

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39 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the planetarium, in the xy-plane, a circle with equation $(x-4)^2+(y-10)^2=25$ represents a boundary at the wildlife sanctuary. What is the greatest value of b for which the horizontal line $y=b$ is tangent to the circle?

a. 15 ✓

b. 5

c. 35

d. 9

The circle has center $(4, 10)$ and radius $r=5$. The highest point on the circle is $(4, 10+5)=(4, 15)$, so the greatest value of b for which $y=b$ is tangent to the circle is 15.

40 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the wind energy cooperative, in the xy-plane, the graph of $y=(x-3)^2-49$ at the orchestra hall intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2-x_1|$?

a. 49

b. 17

c. 14 ✓

d. 7

Setting $y=0$ gives $(x-3)^2=49 \Rightarrow x-3=\pm 7$, so the x-intercepts are -4 and 10. The distance between them is $(10)-(-4)=14$.

41 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In a model for the marine research vessel, in the xy-plane, a circle with equation $(x-7)^2+(y-6)^2=64$ represents a boundary at the sports complex. What is the greatest value of b for which the horizontal line $y=b$ is tangent to the circle?

a. 15

b. 14 ✓

c. -2

d. 70

The circle has center $(7, 6)$ and radius $r=8$. The highest point on the circle is $(7, 6+8)=(7, 14)$, so the greatest value of b for which $y=b$ is tangent to the circle is 14.

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42 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the graph of $y = (x - 3)^2 - 49$ at the orchestra hall intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

a. 49

b. 17

c. 14 ✓

d. 7

Setting $y = 0$ gives $(x - 3)^2 = 49 \Rightarrow x - 3 = \pm 7$, so the x-intercepts are -4 and 10. The distance between them is $(10) - (-4) = 14$.

43 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, a circle with equation $(x - 7)^2 + (y - 6)^2 = 64$ represents a boundary at the sports complex. What is the greatest value of b for which the horizontal line $y = b$ is tangent to the circle?

a. 15

b. 14 ✓

c. -2

d. 70

The circle has center $(7, 6)$ and radius $r = 8$. The highest point on the circle is $(7, 6 + 8) = (7, 14)$, so the greatest value of b for which $y = b$ is tangent to the circle is 14.

44 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane for the charter ferry service model, the parabola $y = x^2 + 12x + k$ and the horizontal line $y = 3$ intersect at exactly one point. What is the value of the constant k?

a. 39 ✓

b. 36

c. 41

d. 42

Setting $x^2 + 12x + k = 3$ gives $x^2 + 12x + (k - 3) = 0$. For the graphs to intersect at exactly one point, the discriminant must equal zero: $(12)^2 - 4(1)(k - 3) = 0 \Rightarrow k - 3 = 36 \Rightarrow k = 39$.

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45 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the graph of $y = (x - 1)^2 - 100$ at the maple syrup farm intersects the x-axis at points $(x_1, 0)$ and $(x_2, 0)$. What is the positive difference $|x_2 - x_1|$?

a. 10

b. 100

c. 21

d. 20 ✓

Setting $y = 0$ gives $(x - 1)^2 = 100 \Rightarrow x - 1 = \pm 10$, so the x-intercepts are -9 and 11. The distance between them is $(11) - (-9) = 20$.

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